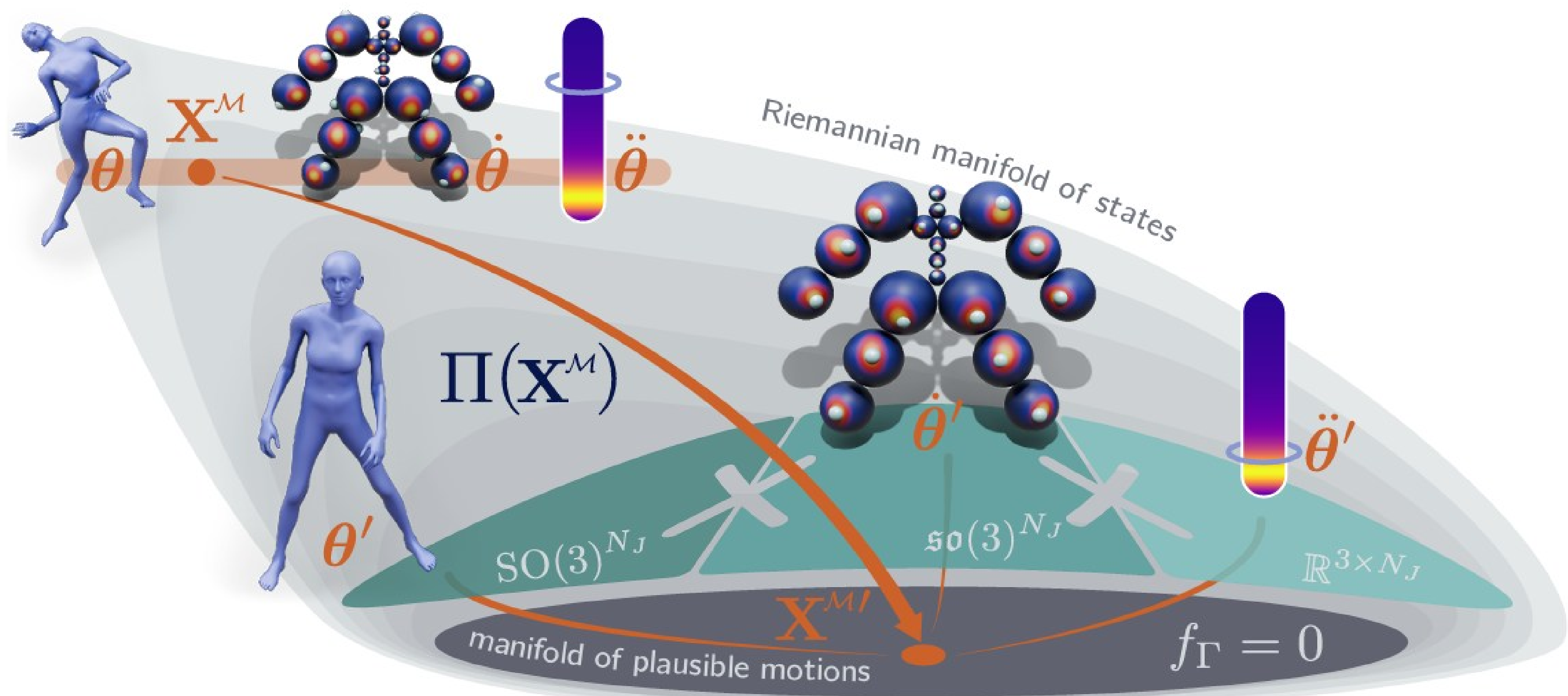
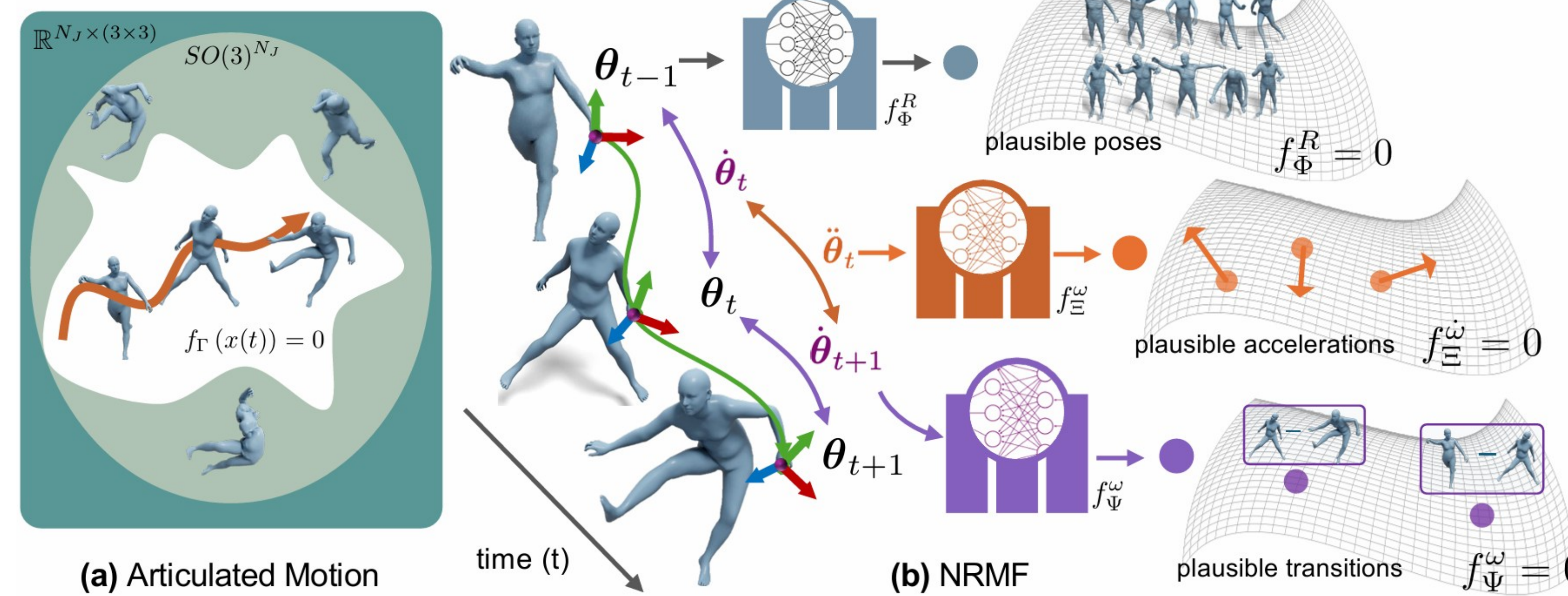


NRMF = Neural Riemannian Motion Fields



NRMF is a general-purpose, expressive and robust unconditional motion prior. It models the space of plausible poses (θ), transitions ($\dot{\theta}$) and accelerations ($\ddot{\theta}$) on the zero-level set of a geometric neural distance field. Poses are depicted along side their transitions and accelerations, which are visualized as blue dots onto the per-joint distributions of learned transitions and as blue rings around the magnitude distribution of all accelerations.

Learning NRMF



(a) Articulated Motion

(b) NRMF

$\mathbf{x} = [t_r, \theta, \dot{\theta}, \ddot{\theta}] \in (\mathbb{R}^3 \times \mathcal{M})$ (Motion state)

$f_\Gamma = \begin{bmatrix} f_\Phi^R(\theta) \\ f_\Psi^\omega(\dot{\theta} | \theta) \\ f_\Xi^\omega(\ddot{\theta} | \theta, \dot{\theta}) \end{bmatrix}$ (pose field) (transition field) (acceleration field) (RMF)

$\theta_{t+1} := \Pi^R(\theta_t) = \text{Exp}_{\theta_t} \left(-\alpha_\theta f_\Phi^R(\theta_t) \frac{\text{grad} f_\Phi^R(\theta_t)}{\|\text{grad} f_\Phi^R(\theta_t)\|} \right)$

$\dot{\theta}_{t+1} := \Pi_t^\omega(\dot{\theta}_t) = \dot{\theta}_t - \alpha_\theta f_\Psi^\omega(\dot{\theta}_t) \frac{\nabla f_\Psi^\omega(\dot{\theta}_t | \theta_t)}{\|\nabla f_\Psi^\omega(\dot{\theta}_t | \theta_t)\|}$

$\ddot{\theta}_{t+1} := \Pi_t^\omega(\ddot{\theta}_t) = \ddot{\theta}_t - \alpha_\omega f_\Xi^\omega(\ddot{\theta}_t) \frac{\nabla f_\Xi^\omega(\ddot{\theta}_t | \theta_t, \dot{\theta}_t)}{\|\nabla f_\Xi^\omega(\ddot{\theta}_t | \theta_t, \dot{\theta}_t)\|}$

α : learning rate, ω : angular velocity, t_r : root translation, $\mathcal{D}$: dataset

$f_\Gamma^* \rightarrow$ Neural Riemannian Motion Field

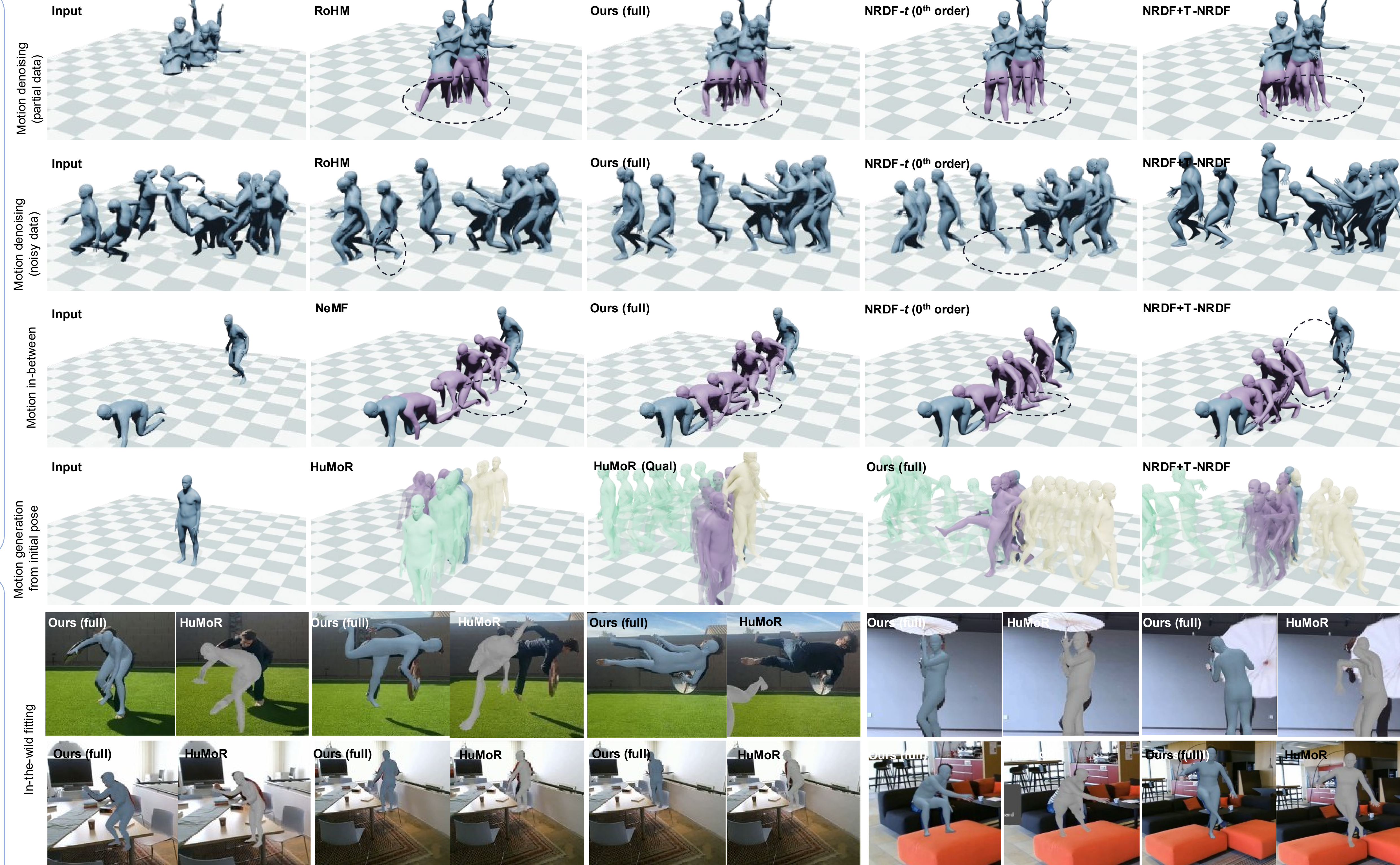


Figure: Qualitative comparisons. The mesh and body parts are the observation inputs and we show the output or optimization results. For in-the-wild fitting, we conduct experiments on both in-door datasets (bottom row) and online videos (upper row). For motion generation, we show samples generated from the common standing pose.

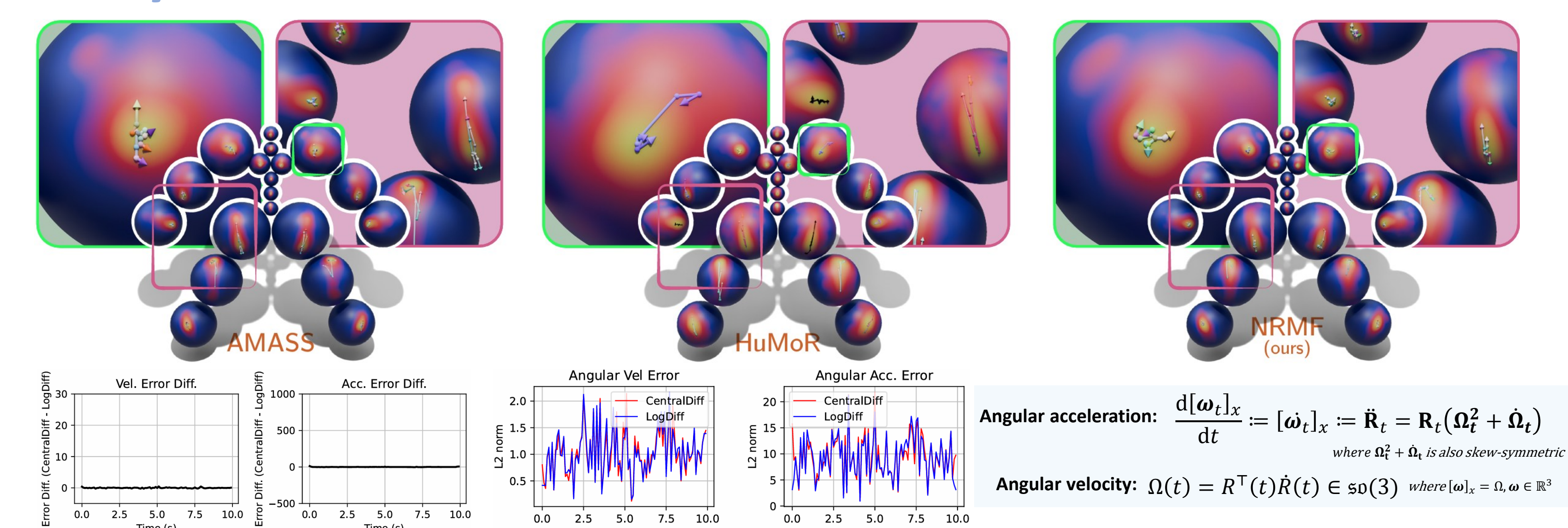
Optimization

$$E_I^*(t_r, \theta, \beta) = \mathcal{L}_{\text{data}} + \lambda_\beta \mathcal{L}_\beta + \lambda_\theta \mathcal{L}_\theta + \lambda_{\text{reg}} \mathcal{L} + \lambda_{\dot{\theta}} \mathcal{L}_{\dot{\theta}} + \lambda_{\ddot{\theta}} \mathcal{L}_{\ddot{\theta}}$$

Algorithm 1 RMF Integrator

Input: An initial pose θ_0 and velocity $\dot{\theta}_0$, a possibly noisy acceleration sequence $\{\ddot{\theta}_t\}_t$ and all components of the trained network f_Γ (and hence the projectors Π),
Output: Plausible motion $\{\mathbf{X}_t\}_t$
1: **for** $t \leftarrow 1, \dots, T$ **do** $\triangleright$ Euler integrate of velocities
2: $\dot{\theta}_t \leftarrow \dot{\theta}_{t-1} + \lambda_t \ddot{\theta}_{t-1}$
3: $\theta_t \leftarrow \Pi^R(\theta_{t-1})$
4: **end for**
5: **for** $t \leftarrow 1, \dots, T$ **do** $\triangleright$ Update poses
6: $\theta_t \leftarrow \text{Exp}_{\theta_{t-1}}(\alpha_t [\dot{\theta}_{t-1}]_x)$
7: $\dot{\theta}_t \leftarrow \Pi^\omega(\dot{\theta}_t)$
8: **end for**
9: Compose $\{\mathbf{X}_t\}_t$ from individual estimates $\{\dot{\theta}_t, \ddot{\theta}_t, \theta_t\}_t$

Analysis



Angular acceleration: $\frac{d[\omega_t]_x}{dt} = [\dot{\omega}_t]_x \equiv \ddot{\mathbf{R}}_t = \mathbf{R}_t(\Omega_t^2 + \Omega_t)$
where $\Omega_t^2 + \Omega_t$ is also skew-symmetric
Angular velocity: $\Omega(t) = \mathbf{R}^T(t)\dot{\mathbf{R}}(t) \in \mathfrak{so}(3)$ where $[\omega]_x = \Omega, \omega \in \mathbb{R}^3$

Evaluation

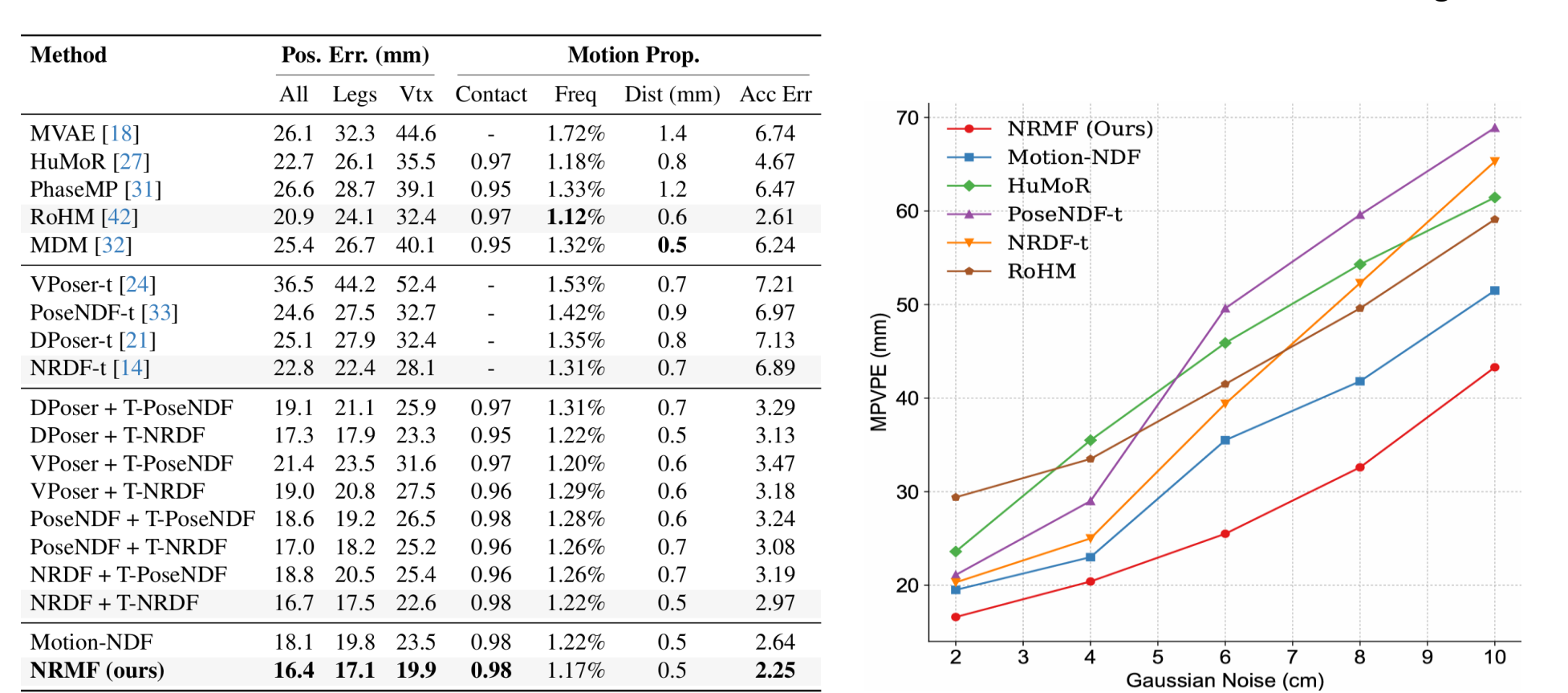
Method	MPJPE (mm)	MPVPE (mm)	Acc Err (m/s^2)	Trans Err ($\times 10^{-3}$)
SMPLer-X [5]	82.65	94.23	23.71	31.63
+ No prior	84.67	96.82	26.75	34.54
+ VPoser [24]	79.98	91.53	25.82	30.58
+ DPoser [21]	75.45	87.04	27.54	28.69
+ PoseNDF [33]	73.49	84.61	25.12	29.77
+ NRDF [14]	71.88	83.23	24.31	26.38
+ T-NRDF	66.98	76.94	9.89	7.98
+ A-NRDF	70.88	81.92	6.73	11.87
+ RoHM [42]	69.78	79.72	9.13	12.37
+ NRMF (full)	66.13	75.61	6.52	5.67

Tab 1: Quantitative evaluation results of motion refinement on 3DPW.

Method	RGB-D	RGB	Method	APD $\uparrow$ (cm)	FID $\downarrow$	FID $\downarrow$	Acc $\downarrow$
	[Acc] Dist FID $\downarrow$ Freq	[Acc] Dist FID $\downarrow$ Freq					
HuMoR [27]	1.88 35.67 11.65 6.13%	2.35 41.92 12.47 10.52%	GMM [41]	16.28	0.435	-	-
PhaseMP [31]	1.79 6.67 12.16 3.21%	1.80 46.96	GAN-S [11]	15.68	0.201	-	-
RoHM [42]	1.79 6.67 12.16 3.21%	2.19 14.31 13.65 4.65%	VPoser [24]	10.75	0.113	-	-
VPoser [24]	3.45 53.49 21.89 10.66%	3.24 54.96 19.64 13.95%	PoseNDF [33]	18.54	1.124	-	-
PoseNDF-t	2.84 48.77 16.45 11.35%	2.93 50.13 16.87 13.46%	DPoser [21]	36.25	2.344	-	-
NRDF-t	2.67 45.91 14.58 9.97%	2.77 46.78 14.32 12.98%	NRDF [14]	24.87	0.706	-	-
DPoser [21]	3.32 49.45 17.32 10.89%	3.08 51.35 16.98 12.21%	MVAE [13]	85.26	1.475	10.571	8.15
NRMF	1.48 5.78 8.95 2.78%	1.73 16.21 14.13 5.12%	HuMoR [27]	98.47	0.278	9.147	4.96
			MotionNDF	93.46	0.894	8.351	-
			NRMF (ours)	96.37	0.718	4.218	3.18

Tab 2: Motion estimation on PROX dataset.

Tab 3: Quantitative evaluation of motion generation.



Tab 4: Motion denoising on AMASS dataset.

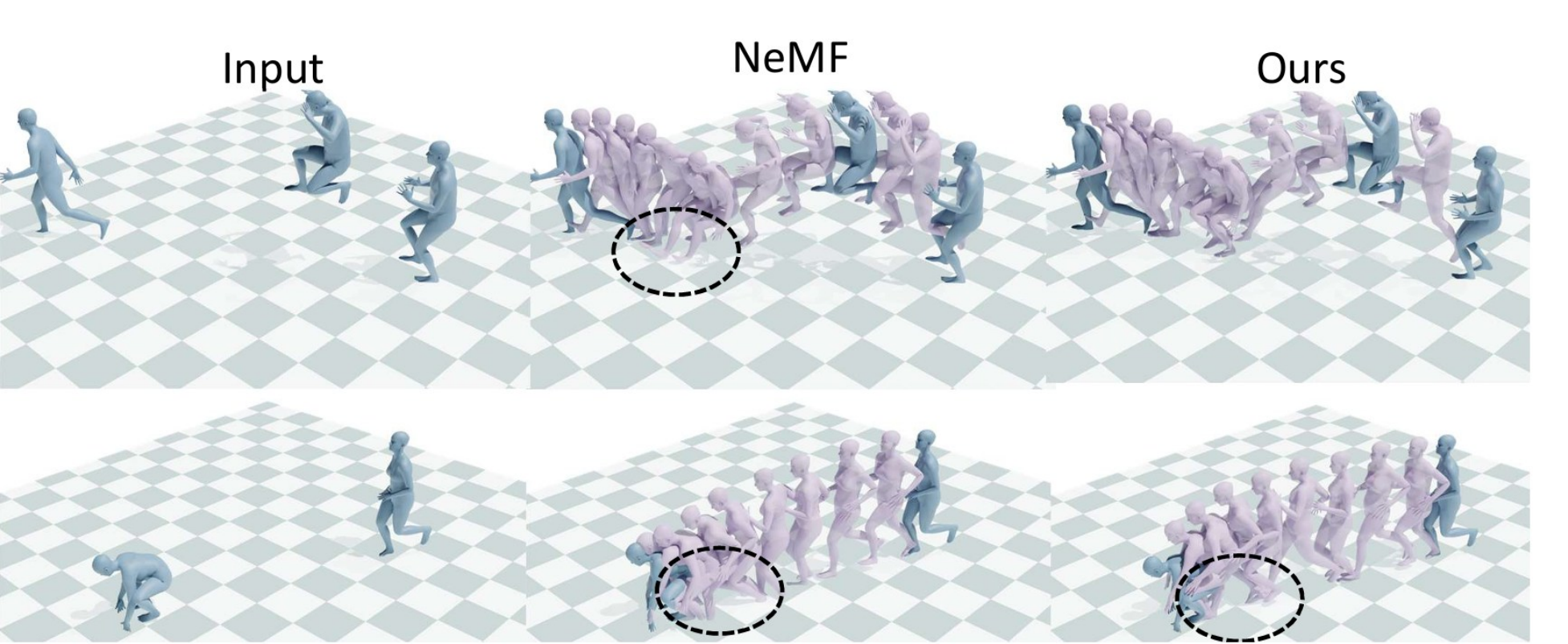
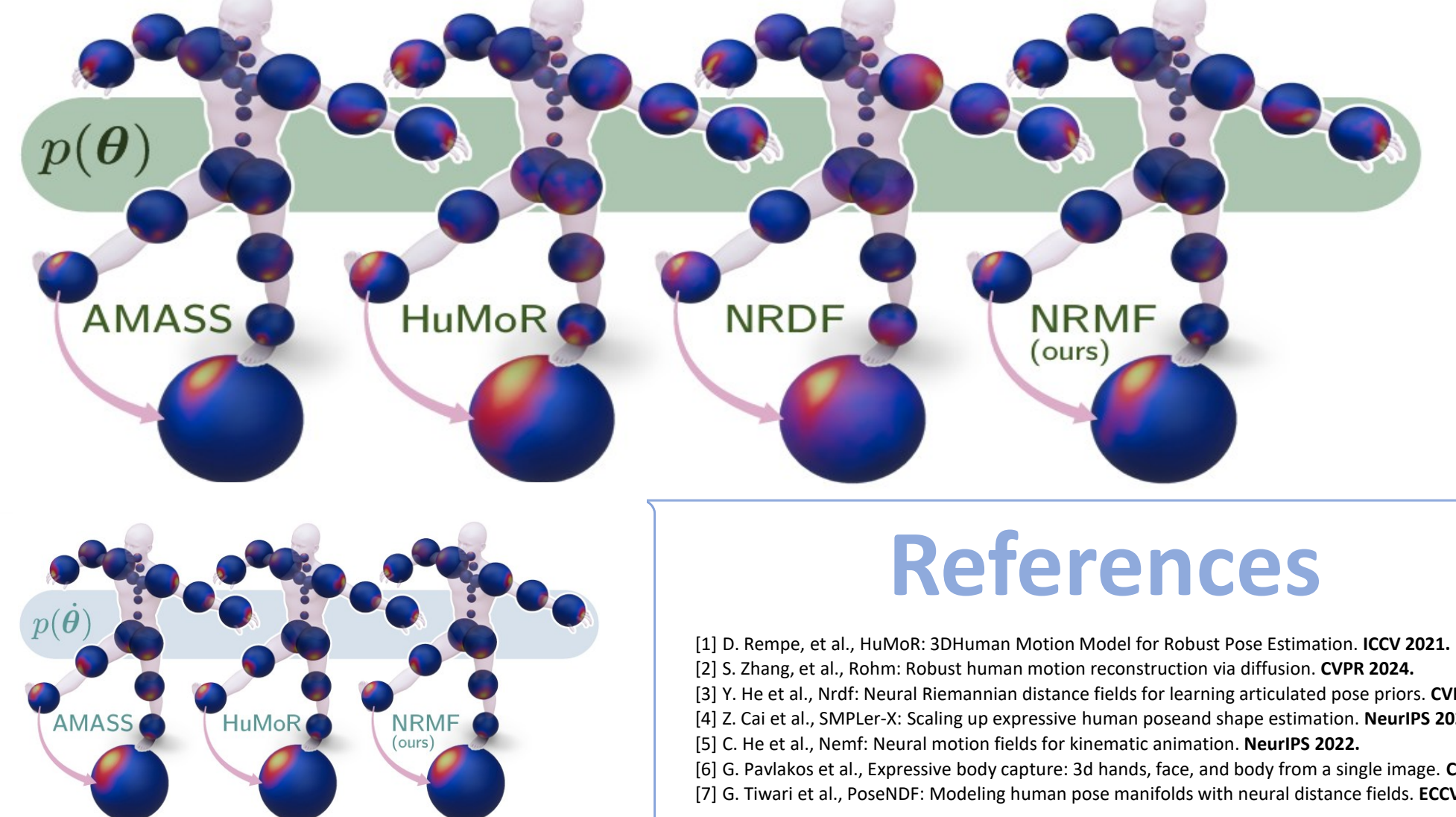


Figure: Comparison on motion in-between.

Distribution



References

- [1] D. Rempel et al., HuMoR: 3D Human Motion Model for Robust Pose Estimation. ICCV 2023.
- [2] S. Zhang et al., Robust: Robust human motion reconstruction via diffusion. CVPR 2024.
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- [5] C. He et al., NeMF: Neural motion fields for kinematic animation. NeurIPS 2022.
- [6] G. Pavlakos et al., Expressive body capture: 3D hands, face, and body from a single image. CVPR 2019.
- [7] G. Tiwari et al., PoseNDF: Modeling human pose manifolds with neural distance fields. ECCV 2022.